

DEMAND SYSTEM ESTIMATION

Using the Almost Ideal Demand System (AIDS)

An Applied Econometrics Portfolio Project in Stata

Yurou Liu

Applied-Econometrics-with-Stata | Research Portfolio

Executive Summary

Objective. Estimate a theoretically coherent system of consumer demand equations and quantify how product expenditure shares respond to own prices, competitors' prices, and total expenditure.

Approach. The analysis constructs budget shares for 24 products observed in 47 geographic markets over two periods (2,256 product-market observations). It estimates a restricted Linear Almost Ideal Demand System (LA-AIDS), addresses price endogeneity with a Hausman-type instrument, compares 2SLS and efficient GMM, iterates the Exact AIDS price index, and translates the preferred estimates into economically interpretable elasticities.

Main findings. The own-price coefficient is negative and the average cross-price coefficient is positive across specifications, consistent with downward-sloping demand and substitution across products. The first-stage F-statistic is 1,754.34, indicating a strong instrument. IV and GMM yield a larger own-price response than OLS, suggesting that price endogeneity attenuates the OLS estimate. Exact AIDS and LA-AIDS produce similar coefficients, supporting the Stone index as a useful approximation in this application. The preferred model implies an own-price elasticity of -2.725 and an expenditure elasticity close to one.

Project at a Glance

Dimension	Portfolio Evidence
Data	2,256 observations; 47 geographic markets; 2 time periods; 24 products
Demand model	Restricted LA-AIDS and iterated Exact AIDS
Identification	Hausman-type cross-market price instrument
Estimators	Cluster-robust OLS, 2SLS, and GMM
Economic outputs	Own-price, cross-price, expenditure, and Hicksian elasticities
Software	Stata; reproducible modular workflow

1. Research Question and Economic Framework

Consumer demand estimation links observed market outcomes to the behavioral responses that matter for pricing, competition, and welfare analysis. The central question is: how does each product's expenditure share change when its own price, competing products' prices, or total expenditure changes?

The Almost Ideal Demand System provides a flexible framework that is consistent with consumer demand theory and yields interpretable price and expenditure elasticities. For product i in market j and period t , the budget-share equation is represented as:

$$w_{ijt} = \alpha_i + \sum_k \gamma_{ik} \ln(p_{kjt}) + \beta_i \ln(X_{jt} / P_{jt}) + \varepsilon_{ijt}$$

To reduce dimensionality while preserving the adding-up restriction, the empirical specification imposes a common cross-price coefficient and recovers the own-price coefficient from:

$$\gamma_{\text{own}} + (K - 1)\gamma_{\text{cross}} = 0$$

This restriction makes the estimated price response internally consistent across the 24-product system and allows the analysis to focus on a compact set of economically meaningful parameters.

2. Data and Variable Construction

The dataset is organized at the product-market-time level. A market identifier combines geographic location and quarter. Expenditure is calculated as price per serving multiplied by servings sold; total market expenditure is the sum across products; and each product's budget share is its expenditure divided by the market total.

2.1 Sample Summary

Metric	Value	Interpretation
Observations	2,256	Product-market-time rows
Geographic markets	47	Distinct locations
Time periods	2	Quarter-level periods
Products	24	Balanced product set
Market-time cells	94	47 markets $\times$ 2 periods

The adding-up condition is verified directly: product budget shares sum to one within each market-time cell. This data check is essential because the AIDS model is defined over a complete allocation of expenditure.

2.2 Budget-Share Distribution

Mean	Std. Dev.	Minimum	Median	Maximum
0.0417	0.0499	0.00047	0.0253	0.4951

The average share is approximately $1/24$, as expected in a 24-product system. The right-skewed distribution indicates that a small number of products account for comparatively large expenditure shares, while many products occupy smaller positions within a market basket.

3. Price Index and LA-AIDS Estimation

3.1 Stone Price Index

The linearized AIDS model replaces the nonlinear exact price index with the Stone price index, a budget-share-weighted average of log prices:

$$\ln(P^s_{jt}) = \sum_i w_{ijt} \ln(p_{ijt})$$

94	-2.085	0.065	[-2.249, -1.928]
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The limited variation in the Stone index across market-time cells helps explain why the linear approximation and the iterated exact index produce similar parameter estimates later in the analysis.

3.2 Restricted LA-AIDS Results

The restricted OLS specification includes product fixed effects and market-clustered standard errors. The model estimates one common cross-price effect and derives the own-price coefficient from the adding-up restriction.

Parameter	Estimate	Economic sign	Interpretation	Model statistic
γ_{cross}	0.00313	Positive	Competitor price increases shift share toward product i	$R^2 = 0.304$
γ_{own}	-0.07188	Negative	Own-price increases reduce product i expenditure share	$N = 2,256$
β	≈ 0	Near zero	Budget shares change little with real expenditure	94 clusters

The signs align with demand theory: own-price effects are negative and cross-price effects are positive. However, OLS may be inconsistent because firms can adjust prices in response to unobserved local demand shocks. The next section addresses this identification problem.

4. Instrumental Variables and Price Endogeneity

4.1 Identification Strategy

The analysis uses a Hausman-type instrument based on the same product's price in other markets. The identifying logic is that common cost shocks generate correlation in prices across markets, while local demand shocks remain geographically specific. Instrument validity therefore requires shared cost variation, market-specific demand shocks, and no direct cross-market pricing response to the local structural error.

4.2 First-Stage Strength

First-stage measure	Estimate	Assessment
Instrument coefficient, π_1	0.523	Positive and precisely estimated
Excluded-instrument F-statistic	1,754.34	Far above conventional weak-instrument thresholds

The first-stage evidence strongly supports instrument relevance. Exogeneity remains an economic assumption rather than something established by the first-stage statistic, so the interpretation depends on the absence of correlated demand shocks across markets.

5. 2SLS and GMM Results

The IV estimates strengthen the magnitude of the negative own-price response relative to OLS. This pattern is consistent with attenuation in the OLS price coefficient when firms increase prices in markets experiencing favorable unobserved demand shocks.

Specification	γ_{cross}	γ_{own}	β	Interpretation
LA-AIDS OLS	0.00313	-0.07188	≈ 0	Baseline; potentially endogenous prices
2SLS	0.003705	-0.085221	-0.000029	Larger price response after IV correction
Efficient GMM	0.003705	-0.085221	-0.000029	Numerically equal to 2SLS in exactly identified model
Exact AIDS	0.00313	-0.07188	≈ 0	Preferred nonlinear price-index benchmark

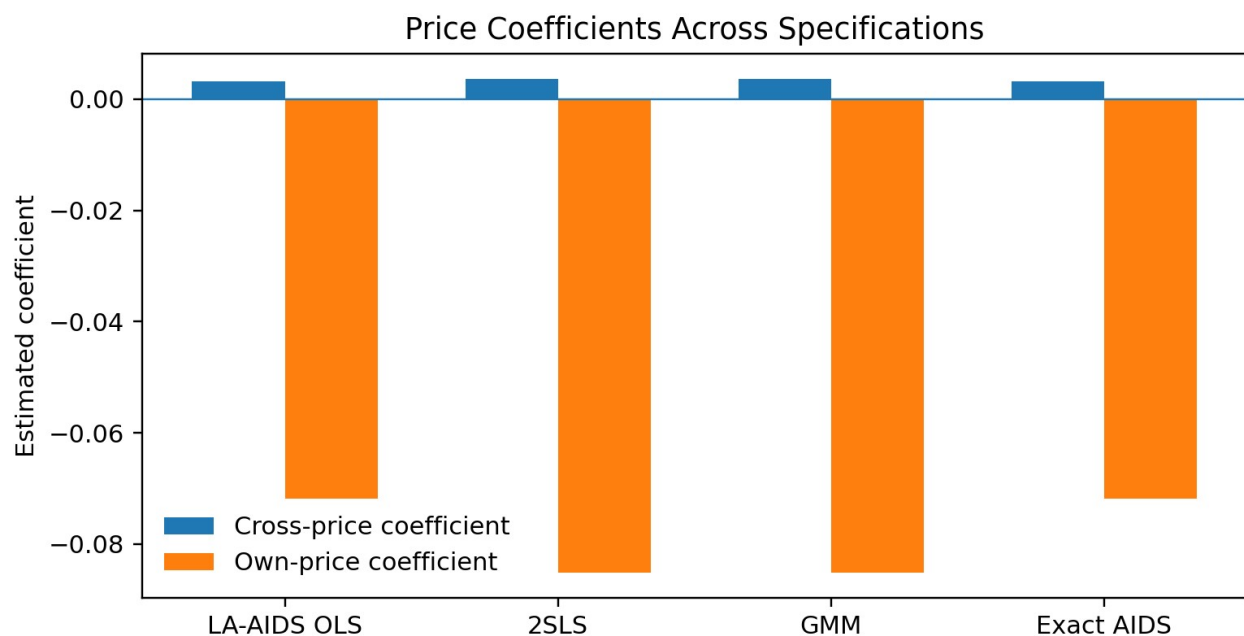


Figure 1. Estimated own- and cross-price coefficients across model specifications.

Why 2SLS and GMM coincide

The model is exactly identified: the number of excluded instruments equals the number of endogenous regressors. Under exact identification, the sample moment conditions pin down the same coefficient vector regardless of the GMM weighting matrix. Efficient GMM therefore reproduces the 2SLS point estimates rather than delivering additional efficiency gains through overidentifying restrictions.

6. Exact AIDS Price Index

The exact AIDS price index depends on the unknown demand parameters, while those parameters can only be estimated after the real-expenditure term has been constructed. This circularity makes the exact model nonlinear. The analysis resolves it through iteration: initialize parameters from LA-AIDS,

construct the exact price index, re-estimate the share equation, update parameters, and repeat until the maximum parameter change is below 10^{-4} .

Convergence output	Value	Interpretation
Iterations	1	Rapid convergence from LA-AIDS starting values
Final γ_{cross}	0.00313	Positive substitution effect
Final γ_{own}	-0.07188	Negative own-price effect
Final β	4.66×10^{-17}	Economically indistinguishable from zero

The Exact AIDS and LA-AIDS estimates are nearly identical. Given moderate price-index variation and relatively stable shares, the Stone index closely approximates the nonlinear price aggregator. This result supports using LA-AIDS when transparency and computational simplicity are priorities, while retaining Exact AIDS as a useful robustness benchmark.

7. Elasticity Analysis

Elasticities translate model coefficients into intuitive behavioral responses. Using the preferred Exact AIDS estimates and the mean budget share of 0.0416667, the analysis calculates Marshallian and Hicksian price responses as well as expenditure responsiveness.

Elasticity	Estimate	Classification	Economic meaning
Marshallian own-price	-2.725	Elastic	A 1% own-price increase is associated with an approximately 2.73% decline in quantity demanded, holding nominal expenditure fixed.
Average cross-price	0.075	Weak substitutes	A 1% increase in another product's price raises demand for product <i>i</i> by about 0.08% on average.
Expenditure	1.000	Unit expenditure elastic	Demand rises approximately proportionally with total expenditure.
Hicksian own-price	-2.683	Elastic compensated response	The substitution effect remains large after compensating consumers for purchasing-power changes.

Elasticity Estimates from the Preferred Exact AIDS Specification

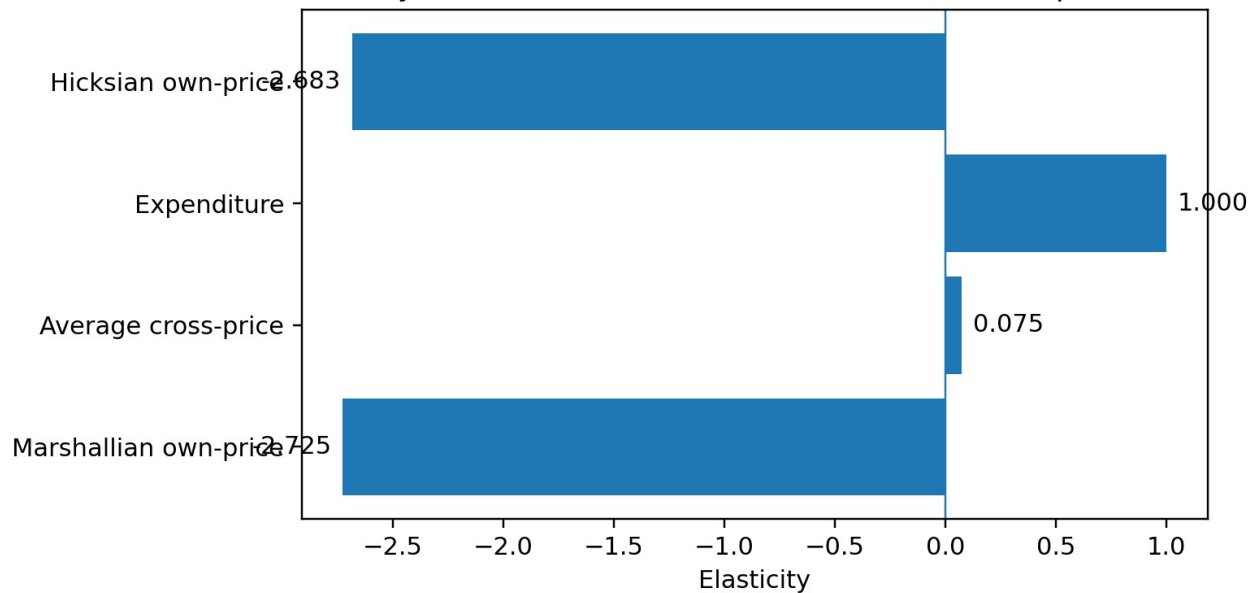


Figure 2. Elasticity estimates from the preferred Exact AIDS specification.

The main economic result is a strong own-price response: both uncompensated and compensated elasticities are below -2.6. The positive but small cross-price elasticity suggests broad substitution across products, but only weak substitution with any single competitor. An expenditure elasticity near one indicates that the representative product behaves neither as a strong necessity nor as a luxury at the sample mean.

8. Interpretation, Limitations, and Extensions

8.1 Interpretation

Across estimators, the demand system delivers stable qualitative conclusions. Consumers reduce demand when a product's own price rises and reallocate a small share of spending toward a product when competitors become more expensive. Correcting for price endogeneity increases the magnitude of the estimated price response, underscoring the importance of separating supply-side price movements from demand shocks.

8.2 Limitations

- The common cross-price restriction imposes symmetry across all product pairs and may mask heterogeneous substitution patterns.
- The Hausman-type instrument relies on cross-market demand shocks being uncorrelated; this exclusion restriction cannot be verified solely from the observed data.
- Elasticities are evaluated at the mean budget share and should be interpreted as representative rather than product-specific responses.

8.3 Natural Extensions

- Estimate product-specific own- and cross-price coefficients under the full AIDS system.
- Test alternative instruments and report overidentification diagnostics in an overidentified specification.
- Extend the model to demographic interactions, market heterogeneity, or panel demand dynamics.

9. Reproducibility and Portfolio Value

The accompanying Stata script implements the workflow from raw product-market data through variable construction, model estimation, iterative Exact AIDS convergence, and elasticity calculation. The repository is structured to separate code from analytical output and to make the empirical sequence easy to audit.

This project demonstrates the following applied-econometrics capabilities:

- Translating consumer theory into an estimable demand system
- Constructing market-level expenditure and price-index variables
- Diagnosing and addressing endogenous regressors
- Implementing cluster-robust OLS, 2SLS, and GMM in Stata
- Designing and validating an iterative nonlinear estimation routine
- Communicating coefficients through economically interpretable elasticities

Conclusion

The project estimates consumer demand using linearized and exact AIDS specifications. Across models, own-price effects are negative, cross-price effects are positive, and correcting for price endogeneity increases the estimated response. The strong first stage, the agreement of 2SLS and GMM under exact identification, and the similarity of LA-AIDS and Exact AIDS together provide a coherent, reproducible applied microeconomic result.